

K L Deemed to be University Department of Electronics and Communication Engineering -- KLVZA Course Handout 2026-2027, Odd Sem	
Course Title	: FUNDAMENTALS OF BIG DATA
Course Code	: OEAD0001
L-T-P-S Structure	: 3-0-0-0
Pre-requisite	: Nil
Credits	: 3
Course Coordinator	: PRAVALLIKA KOPPULA
Team of Instructors	: Gnanadeepthi Bitra, PRAVALLIKA KOPPULA
Teaching Associates	:
Syllabus :	Module 1: Introduction to Big Data Analytics – Big Data overview, analytics in industry verticals, data analytics lifecycle (discovery, data preparation, model planning, model building, communicating results, operationalizing), initial data exploration, basic data visualization, summary statistics Module 2: Advanced Analytics and Statistical Modeling – Selecting analytics techniques based on business objectives, common analytics algorithms (regression, clustering), use-case suitability of techniques, model validation and diagnostics. Module 3: Big Data Technology & Tools – MapReduce programming paradigm, Hadoop, HDFS, Hive for analytics. Module 4: Apache Spark, Operationalizing Analytics Projects with PIG – Applying Pig-database analytics, steps to operationalize analytics projects, deliverables in analytics lifecycle, presentation creation for stakeholders.
Text Books :	1. Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data by EMC Education Services 2014 2. David Dietrich, Barry Heller, Beibei Yang, “Data Science and Big Data Analytics,” EMC Education Series, John Wiley, ISBN: 978-1-118-87613-8, 2015. 3. Peter Bühlmann, Petros Drineas, Michael Kane, Mark van der Laan, “Handbook of Big Data”, CRC Press, 2016.
Reference Books :	1 Big Data: A Revolution That Will Transform How We Live, Work, and Think Viktor Mayer Schonberger and Kenneth Cukier Atlantic Publishers 2013 2 Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking Foster Provost and Tom Fawcett O'Reilly Media 2013 3 Hadoop: The Definitive Guide Tom White O'Reilly Media 2015 4 Data Analytics Made Accessible Anil Maheshwari Apress 2023 5 Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph David Loshin Morgan Kaufmann 2013
Web Links :	1. https://www.edx.org/course/big-data-analytics-adelaide-analyticsx 2. https://www.coursera.org/specializations/big-data
MOOCS :	1. https://www.coursera.org/learn/big-data-management?action=enroll&specialization=big-data 2. https://www.coursera.org/learn/big-data-machine-learning?specialization=big-data
Course Rationale :	Students will gain knowledge on analyzing Big Data. It serves as an introductory course for graduate students who are expecting to face Big Data storage, processing, analysis, visualization, and application issues on both workplaces and research environments.
Course Objectives :	The objective of the Big Data analytics and Web Engineer course enables students understands the overview of data capturing through Embedded Systems and IoT devices, fundamentals of ‘R,’ Map Reduce, Hadoop and to Calibrate Big data Analytical tools for business intelligence applications.
Industry Specific Tools :	1. Hadoop 2. HDFS 3. Hive 4. Pig 5. Apache Spark

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Illustrate the concepts of big data, Initial exploration of analysis of data and Data visualization.	PO1,PSO2	2
CO2	Implement Initial exploration of data and advanced data analytics by using R	PO2,PO3,PSO2	3
CO3	Apply advanced algorithms & Statistical modeling for big data using HDFS, HIVE.	PO3,PO4,PSO2	3
CO4	Apply advanced SQL functions and Pig tools for common deliverables of analytics life cycle project.	PO3,PO4,PSO2	3

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-2	COI-3
CO1	2	Btl-2 Undersatnd the concepts of big data, Data visualization Concepts	
CO2	3	Btl-2 Undersatand the concepts of data exploration and advanced data analytics by using R	Btl-3 Implement Initial exploration of data and advanced data analytics by using R
CO3	3	Btl-2 understand advanced algorithms & Statistical modeling for big data using HDFS and HIVE.	Btl-3 Apply advanced algorithms & Statistical modeling for big data using HDFS and HIVE
CO4	3	Btl-2 Undersatnd the concepts of advanced SQL functions, HIVE along with common deliverables of analytics life cycle project.	Btl-3 Apply advanced SQL functions for in database analytics by HIVE along with common deliverables of analytics life cycle project

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
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Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Introduction to Big Data Analytics:	T BOOK [1], CH2 Page no 26-52	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
2	CO1	COI-2	Big Data overview, analytics in industry verticals	T BOOK [1], CH2 Page no 26-52	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
3	CO1	COI-2	Overview of Data Analytics, Data Analytics Life cycle	T BOOK [1], CH2, Page no 53-57	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1
4	CO1	COI-2	data Discovery, Data Preparation, Model Planning	T BOOK [1], CH2, Page no 53-57	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
5	CO1	COI-2	Model Building Communicating Results and Findings, Operationalizing	TBOOK [1], CH2, Page no 53-57	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
6	CO1	COI-2	Initial Analysis of the Data: Initial Exploration	T BOOK [4], CH2, Page no 53-57	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1
7	CO1	COI-2	Apply basic analytics methods such as distributions	T BOOK [1],CH3,Page no 102,104,111	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
8	CO1	COI-2	statistical tests and summary operations, and differentiate between results that are statistically	T BOOK [1], CH3 Page no 112	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			sound vs statistically significant			
9	CO1	COI-2	Identify a model for your data and define the null and alternative hypothesis T BOOK [1], CH3 Page no112 Chalk,LTC,PPT,Talk End Semester Exam,Home Assignment,SEM EXAM1 10 CO1COI-2 Experimenta	T BOOK [1], CH3 Page no11	Chalk,LTC,PPT,Talk	End Semester Exam,Home Assignment,SEM-EXAM1
10	CO1	COI-2	Experimentation and demonstration of initial analysis of data using R	T BOOK [1], CH3 Page no112	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
11	CO1	COI-2	Basic data analytics, reporting, and apply basic data visualization techniques to your data	T BOOK [1],CH3,Page no102,104,111	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
12	CO2	COI-2	Need of analytic and select an appropriate technique based on business objectives	TBOOK[1],CH3.2.4,Page no80113	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
13	CO2	COI-2	initial hypotheses; and the data's structure and volume	T BOOK[1],CH3.2.4,Page no 80113	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1
14	CO2	COI-3	Apply some of the more methods in Analytics solutions, algorithms and the technical foundations for the methods	T BOOK [1],CH4, Page no 117	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
15	CO2	COI-3	The environment (use case) in which each technique can provide the most value	T BOOK [1], CH4.2, Page no120,123	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
16	CO2	COI-3	Use appropriate diagnostic methods to validate the models created Use R	T BOOK [1], CH5, Page no 137	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1
17	CO2	COI-3	Database analytical functions to fit,	T BOOK [1], CH5, Page no 137	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			score and evaluate models			
18	CO2	COI-3	models created Use R and indatabase analytical functions to fit,	T BOOK [1], CH5.5.1 Page no146	Chalk,LTC,PPT,Talk	End Semester Exam,Home Assignment,SEM-EXAM1
19	CO3	COI-2	Data — Technology & Tools: Tool to Perform Analytics on Unstructured data using MapReduce Programming paradigm	T BOOK [1], CH10, Pageno295,300,308	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
20	CO3	COI-3	Use Hadoop: Hadoop Architecture and Components Implementing Distributed Data Processing Using Hadoop	T BOOK [1], CH10,Pageno295,300,308	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
21	CO3	COI-3	HDFS (Hadoop Distributed File System) HDFS Architecture and Components, HDFS Operations and Data Management	T BOOK [1], CH10, Pageno296,301	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
22	CO3	COI-3	HIVE: Hive Architecture and Data Models, Hive Query Language (HiveQL) and Data Processing	T BOOK [1], CH10, PageNo295,300,30	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
23	CO3	COI-3	PIG, products in the Hadoop ecosystem other products in the Hadoop ecosystem for unstructured data analytics	T BOOK [1] CH10.2Pageno323,328	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
24	CO3	COI-3	Effectively use advanced SQL functions and Greenplum extensions for in database analytics	T BOOK [1] CH10.2Pageno323,328	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
25	CO3	COI-3	use advanced SQL functions and Greenplum extensions for in database analytics	T BOOK [1] CH11.1.3 Pageno334	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
26	CO3	COI-3	Use MADlib to solve analytics	T BOOK[1]CH11.3.4 Page no 356,357	Chalk,LTC,PPT,Talk	End Semester Exam,Home

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			problems in database			Assignment,SEM-EXAM2
27	CO3	COI-3	MADLib to Solve Analytics Problems in Database	TBOOK[1]CH11.3.4Pageno356,357	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
28	CO4	COI-2	Endgame Operationalizing an Analytics Project: Tasks needed to operationalize an analytics project	T BOOK [1],CH12,Page no 359	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
29	CO4	COI-2	Four common deliverables of an analytics lifecycle project meet the needs of key stakeholders	T BOOK [1],CH12,Page no 359	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
30	CO4	COI-3	Use a framework for creating final presentations for sponsors and analysts.	T BOOK [1],CH12,Page no 359	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
31	CO4	COI-3	Evaluate data visualization and identify ways to improve it	T BOOK [1],CH12, Page no 360	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
32	CO4	COI-3	Apply these concepts to a big data analytics problem in the final lab	T BOOK [1], CH 12 Pageno370,377	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
33	CO4	COI-3	Apply these Spark concepts to a big data analytics	T BOOK [1], CH 12 Pageno370,377	Chalk,LTC,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
34	CO4	COI-2	Evaluate the effectiveness of stream processing tools like Apache Kafka and Spark Streaming in handling real-time data workflows.	T BOOK [1], CH 12 Page no370,377	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
35	CO4	COI-3	Evaluate the effectiveness of stream processing tools like Apache Kafka and Spark Streaming in handling real-time data workflows	T BOOK [1], CH 12 Pageno370,377	Chalk,LTC,PPT,Talk	End Semester Exam,Home Assignment,SEM-EXAM2
36	CO4	COI-2	scalability and efficiency of distributed machine	BOOK [1], CH 12 Page no370,377	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			learning frameworks like Spark MLlib or TensorFlowOnSpark for handling large datasets			
37	CO4	COI-3	scalability and efficiency of distributed machine learning frameworks like Spark MLlib or TensorFlowOnSpark for handling large datasets.	T BOOK [1], CH 12 Page no 370,377	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
38	CO4	COI-3	Evaluate the implementation of data masking and anonymization techniques in securing sensitive data within large datasets lab	TBOOK [1],CH 12, Page no 360	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
39	CO4	COI-2	Revision of CO3 & CO4	T BOOK [1]	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
40	CO4	COI-2	Course conclusion	T Book[1][2]	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Hand out explanation and introduction to Big data Analytics

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE --
20	Handout Explanation	2	PPT	--- NOT APPLICABLE --
20	Introduction to Big Data	2	PPT	--- NOT APPLICABLE --
5	summary of the session	2	PPT	--- NOT APPLICABLE --

SESSION NUMBER : 2

Session Outcome: 1 Student able to understand Overview of Data Analytics Life cycle, Discovery

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	State of the Practice of Analytics	2	PPT	--- NOT APPLICABLE -- -
20	Big Data Analytics in Industry Verticals	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 3

Session Outcome: 1 Students can able to understand and explain the complete Data Analytics Life Cycle, including its phases, objectives, and significance in solving real-world business problems

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Overview of Data Analytics	2	PPT	One minute paper
20	Data Analytics Life cycle	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 4

Session Outcome: 1 Students can able to explain and relate the Discovery, Data Preparation, and Model Planning phases for developing effective data analytics solutions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Data Discovery, Data Preparation,	2	PPT	--- NOT APPLICABLE -- -
20	Model Planning	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 5

Session Outcome: 1 1 Students can able to understanding of model development, result interpretation, stakeholder communication, and deployment of analytics solutions in real-world scenarios

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Model Building Communicating Results and Findings	2	PPT	--- NOT APPLICABLE -- -
20	OperationalizinG	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 6

Session Outcome: 1 Students can able to explore, summarize, and interpret datasets to identify patterns, data quality issues, and meaningful insights for further analytical processing.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Initial Analysis of the Data	2	PPT	Problem-Based Learning
20	Initial Exploration	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 7

Session Outcome: 1 Students will be able to understand and apply basic analytical methods such as frequency distributions and probability distributions to summarize and interpret data effectively.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Apply basic analytics methods	2	PPT	--- NOT APPLICABLE -- -
20	Apply analytics methods- distributions	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 8

Session Outcome: 1 Students will be able to create differentiate between statistically sound results and statistically significant outcomes for effective data-driven decision making

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Statistical tests and summary operations	2	PPT	--- NOT APPLICABLE -- -
20	Differentiate between results that are statistically sound vs statistically significant	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 9

Session Outcome: 1 Students will be able to identify an appropriate statistical model for a dataset and formulate null and alternative hypotheses to perform hypothesis testing and validate analytical findings

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Identify a model for your data	2	PPT	--- NOT APPLICABLE -- -
20	define the null and alternative hypothesis	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 10

Session Outcome: 1 Students will be able to perform initial data analysis using R, apply exploratory techniques, and interpret significant patterns and insights from datasets.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Experimentation and demonstration of initial analysis of data using R	2	PPT	--- NOT APPLICABLE -- -
20	case study demonstration of initial analysis of data using R	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 11

Session Outcome: 1 1 Students will be able to perform basic data analytics, generate reports, and apply data visualization techniques to effectively communicate significant insights from datasets1

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Basic data analytics, reporting,	2	PPT	--- NOT APPLICABLE -- -
20	apply basic data visualization techniques to your data.	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 12

Session Outcome: 1 Students will be able to identify the need for analytics and select appropriate analytical techniques based on business objectives to derive meaningful and significant insights for decision-making

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Need of analytic	2	PPT	--- NOT APPLICABLE -- -
20	select an appropriate technique based on business objectives	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 13

Session Outcome: 1 1 Students will be able to formulate initial hypotheses and analyze the structure and volume of data to identify significant patterns, trends, and analytical opportunities

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	PPT	--- NOT APPLICABLE -- -
20	Initial Analysis of the Data	2	PPT	Active Listening
20	data's structure and volume	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 14

Session Outcome: 1 Students will be able to apply advanced analytics methods and algorithms, and understand their technical foundations to derive significant insights and support effective decision-making.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Apply some of the more methods in Analytics solutions,	3	PPT	--- NOT APPLICABLE -- -
20	Algorithms and the technical foundations for the methods	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 15

Session Outcome: 1 Students will be able to evaluate the environments and use cases in which different analytical techniques provide the most significant value for solving business and data-driven problems

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	The environment (use case) in which each technique can provide the most value	3	PPT	--- NOT APPLICABLE -- -
20	use case of each techniques	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 16

Session Outcome: 1 Students will be able to use appropriate diagnostic methods in R to validate analytical models and assess their performance, reliability, and statistical significance.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Use appropriate diagnostic methods to validate the models created Use R	3	PPT	Case Study
20	Validating the models created Use R	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 17

Session Outcome: 1 Students will be able to apply database analytical functions to fit, score, and evaluate models, and interpret significant results to support data-driven decision making

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Database analytical functions to fit	3	PPT	--- NOT APPLICABLE -- -
20	score & evaluate models	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 18

Session Outcome: 1 Students will be able to use R and in-database analytical functions to fit, score, and evaluate predictive models

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	PPT	--- NOT APPLICABLE -- -
20	Model Building Using R	3	PPT	--- NOT APPLICABLE -- -
20	In-Database Analytics Concepts-Model Fitting	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 19

Session Outcome: 1 1 Students will be able to use analytics tools and the MapReduce programming paradigm to process and analyze unstructured data, extracting significant insights from large-scale datasets

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	PPT	--- NOT APPLICABLE -- -
20	Data — Technology & Tools: Tool to Perform Analytics on Unstructured data using MapReduce Programming paradigm	2	PPT	--- NOT APPLICABLE -- -
20	MapReduce programming paradigm	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 20

Session Outcome: 1 Students will be able to explain the Hadoop architecture and its core components, including HDFS, MapReduce, and YARN, and their roles in managing and processing large-scale data efficiently.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Use Hadoop: Hadoop Architecture and Components	3	PPT	--- NOT APPLICABLE -- -
20	Implementing Distributed Data Processing Using Hadoop	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 21

Session Outcome: 1 Students will be able to explain the architecture and working of HDFS (Hadoop Distributed File System) and utilize its features for reliable, scalable, and distributed storage of large-scale data.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	HDFS Architecture and Components	3	PPT	Statement- Opinion- Summary
20	HDFS Operations and Data Management	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 22

Session Outcome: 1 Student can able to Understand the architecture and data organization mechanisms of Hive & Apply HiveQL commands to store, manage, and analyze large datasets in Hadoop.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	HIVE: Hive Architecture and Data Models	3	PPT	--- NOT APPLICABLE -- -
20	Hive Query Language (HiveQL) and Data Processing	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 23

Session Outcome: 1 Students will be able to explain the role of Pig and other Hadoop ecosystem tools in processing and analyzing unstructured data

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	PIG, products in the Hadoop ecosystem	3	PPT	--- NOT APPLICABLE -- -
20	Hadoop ecosystem for unstructured data analytics	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 24

Session Outcome: 1 Students will be able to effectively use advanced SQL functions and Greenplum extensions to perform in database analytics, process large datasets

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Effectively use advanced SQL functions	3	PPT	One minute paper
20	Greenplum extensions for in database analytics	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 25

Session Outcome: 1 Students will be able to apply advanced SQL functions and Greenplum extensions to perform efficient in database analytics and analyze large-scale datasets

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	use advanced SQL functions	3	PPT	--- NOT APPLICABLE -- -
20	Greenplum extensions for in database analytics	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 26

Session Outcome: 1 Students will be able to use MADlib in a database environment to implement analytical algorithms and solve data driven problems efficiently.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Introduction to MADlib and In-Database Analytics	3	PPT	--- NOT APPLICABLE -- -
20	MADlib and In-Database Analytics	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 27

Session Outcome: 1 Students will be able to implement and evaluate analytical models using MADlib functions within PostgreSQL/Greenplum databases.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Implementing Analytics Models Using MADlib	3	PPT	--- NOT APPLICABLE -- -
20	Implementing Analytics Models Using MADlib examples	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 28

Session Outcome: 1 1 Students will be able to operationalize analytics projects by deploying, monitoring, and maintaining analytical solutions effectively.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Endgame Operationalizing an Analytics Project	2	PPT	--- NOT APPLICABLE -- -
20	Tasks needed to operationalize an analytics project	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 29

Session Outcome: 1 1 Students will be able to identify the four common deliverables of an analytics lifecycle project and explain how they address the needs of key stakeholders.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Four common deliverables of an analytics lifecycle	2	PPT	--- NOT APPLICABLE -- -
20	analytics lifecycle project meet the needs of key stakeholders	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 30

Session Outcome: 1 Students will be able to use a structured framework to create effective final presentations that communicate analytical findings to sponsors and analysts.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Use a framework for creating final presentations for sponsors and analysts.	3	PPT	Group Discussion
20	Use a framework for creating final presentations for sponsors and analysts.	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 31

Session Outcome: 1 Students will be able to evaluate data visualizations and recommend improvements to enhance clarity, effectiveness, and communication of significant insights

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Evaluate data visualization	3	PPT	--- NOT APPLICABLE -- -
20	identify ways to improve it	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 32

Session Outcome: 1 Students will be able to apply big data analytics concepts, tools, and techniques to solve a real-world analytics problem and effectively interpret the results in the final lab

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Apply Advance analytics concepts to a big data analytics problem in the final lab	3	PPT	--- NOT APPLICABLE -- -
20	Real world examples final lab	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 33

Session Outcome: 1 Students will be able to apply Apache Spark concepts and techniques to process, analyze, and derive meaningful insights from big data analytics problems.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Apply these Spark concepts to a big data analytics	3	PPT	Case Study Based Learning
20	Case study -big data analytics	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 34

Session Outcome: 1 Students will be able to evaluate the effectiveness of Apache Kafka and Spark Streaming for processing real time data workflows

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Fundamentals of Real-Time Data Processing and Stream Processing Architectures	2	PPT	--- NOT APPLICABLE -- -
20	Apache Kafka Architecture and Data Ingestion	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 35

Session Outcome: 1 Students will be able to understand Spark Streaming for processing real-time data workflows based on performance, scalability, reliability, and analytical requirements

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	Apache Spark Streaming for Real-Time Analytics	3	PPT	--- NOT APPLICABLE -- -
20	Performance Evaluation of Kafka and Spark Streaming	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 36

Session Outcome: 1 Students will be able to evaluate the scalability and efficiency of distributed machine learning frameworks such as Spark MLlib

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Introduction to Distributed Machine Learning	2	PPT	--- NOT APPLICABLE -- -
20	Spark MLlib for Large-Scale Machine Learning	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 37

Session Outcome: 1 Students will be able to understand TensorFlowOnSpark for processing large datasets and developing high performance analytics solutions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	3	Talk	--- NOT APPLICABLE -- -
20	TensorFlowOnSpark for Distributed Deep Learnin	3	PPT	--- NOT APPLICABLE -- -
20	Evaluating Scalability and Efficiency	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 38

Session Outcome: 1 Students will be able to implement and evaluate data masking and anonymization techniques to secure sensitive information while maintaining data usability in large datasets

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	A	3	Talk	--- NOT APPLICABLE -- -
20	Implementation of Data Masking Techniques for Sensitive Data Protection	3	PPT	--- NOT APPLICABLE -- -
20	Implementation and Evaluation of Data Anonymization Techniques	3	PPT	--- NOT APPLICABLE -- -
5	summary of the session	3	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 39

Session Outcome: 1 Students will be able to apply and evaluate advanced analytics models & analytics solutions using distributed computing frameworks and real-time data technologies.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Revision of CO3	2	PPT	--- NOT APPLICABLE -- -
20	Revision of CO4	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

SESSION NUMBER : 40

Session Outcome: 1 Students will be able to apply analytics, machine learning, and big data technologies to extract insights and solve real-world data-driven problems effectively

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	2	Talk	--- NOT APPLICABLE -- -
20	Course conclusion	2	PPT	--- NOT APPLICABLE -- -
20	Course conclusion	2	PPT	--- NOT APPLICABLE -- -
5	summary of the session	2	PPT	--- NOT APPLICABLE -- -

Tutorial Course DELIVERY Plan:
NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

NO Delivery Plan Exists

Practical Session wise Teaching – Learning Plan

No Session Plans Exists

Skilling Course DELIVERY Plan:

NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
3	Weekly Homework Assignments	1	Identify a model for your data and define the null and alternative hypothesis	Critically analyze the differences between statistically sound results and statistically significant results in the context of big data analytics. Provide a practical example using R to demonstrate how these differences can influence decision making.	CO1
7	Weekly Homework Assignments	2	models created Use R and in database analytical functions to fit,	This topic focuses on building predictive and analytical models using R programming and in database analytical functions. Students learn how to prepare data, fit statistical and machine learning models, generate predictions (scoring), and evaluate model performance using various metrics. The topic also introduces in database analytics, where computations are performed directly within the database to improve efficiency, scalability, and processing speed for large datasets. By integrating R with database analytical functions, learners gain practical experience in developing and validating data-driven solutions for real-world business problems.	CO2
9	Weekly Homework Assignments	3	PIG, products in the Hadoop ecosystem other products in the Hadoop ecosystem for unstructured data analytics	This topic introduces Apache Pig and other key tools within the Hadoop ecosystem that support the storage, processing, and analysis of large volumes of unstructured and semi structured data. Students learn how Apache Pig simplifies complex MapReduce programming through its high-level scripting language, Pig Latin, enabling efficient data transformation and analysis. The topic also covers Hadoop	CO3

Week	Assignment Type	Assignment No	Topic	Details	co
				ecosystem components such as Hive, HBase, Sqoop, Flume, Oozie, Spark, and ZooKeeper, highlighting their roles in data ingestion, storage, processing, workflow management, and real-time analytics. Learners gain an understanding of how these tools work together to build scalable big data analytics solutions for handling diverse and large-scale datasets	
12	Weekly Homework Assignments	4	Evaluate the implementation of data masking and anonymization techniques in securing sensitive data within large datasets lab	This topic focuses on implementing and evaluating data masking and data anonymization techniques to protect sensitive information in large datasets. Students learn how to conceal confidential data such as personal identifiers, financial details, and healthcare records while preserving data usability for analysis. The lab covers techniques such as static and dynamic data masking, pseudonymization, generalization, and suppression, and assesses their effectiveness in maintaining privacy, security, and compliance with data protection regulations. Through hands-on exercises, learners gain practical experience in securing data while supporting analytical and business requirements	CO4

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Tue	Theory	V-S32,V-S33	V-S32,V-S33	---	---	---	---	---	---	---
	Tutorial	--	--	---	---	---	---	---	---	---
	Lab	--	--	---	---	---	---	---	---	---
	Skilling	--	--	---	---	---	---	---	---	---
Wed	Theory	V-S83,V-S84	V-S83,V-S84	---	---	---	---	---	---	---
	Tutorial	--	--	---	---	---	---	---	---	---
	Lab	--	--	---	---	---	---	---	---	---
	Skilling	--	--	---	---	---	---	---	---	---
Thu	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Fri	Theory	---	V-S32,V-S33	---	---	---	---	---	---	---
	Tutorial	---	--	---	---	---	---	---	---	---

	Hour	1	2	3	4	5	6	7	8	9
	Lab	---	--	---	---	---	---	---	---	---
	Skilling	---	--	---	---	---	---	---	---	---
Sat	Theory	V-S83,V-S84	---	---	---	---	---	---	---	---
	Tutorial	--	---	---	---	---	---	---	---	---
	Lab	--	---	---	---	---	---	---	---	---
	Skilling	--	---	---	---	---	---	---	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4
End Semester Summative Evaluation Total= 40 %	End Semester Exam	Weightage	40		180	10	10	10	10
		Max Marks	100			25	25	25	25
In Semester Formative Evaluation Total= 25 %	Home Assignment and Textbook	Weightage	10		90	2.5	2.5	2.5	2.5
		Max Marks	40			10	10	10	10
	ALM	Weightage	15		90	3.75	3.75	3.75	3.75
		Max Marks	40			10	10	10	10
In Semester Summative Evaluation Total= 35 %	Semester in Exam-II	Weightage	17.5		90			8.75	8.75
		Max Marks	50					25	25
	Semester in Exam-I	Weightage	17.5		90	8.75	8.75		
		Max Marks	50			25	25		

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course

In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Gnanadeepthi Bitra	L	33-MA,84-MA	-	-	-	-
PRAVALLIKA KOPPULA	L	32-MA,83-MA	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(PRAVALLIKA KOPPULA)(9255)(CSE-2)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of AI&DS

HEAD OF DEPARTMENT:**Approval from: DEAN-ACADEMICS**

(Sign with Office Seal)

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